

SYSTEM FOUNDRY / BUSINESS MVP

GearLedger

A local-first return queue that keeps shared equipment, deadlines, condition, and value visible.

PRODUCT STATUS	AUTOMATED TESTS	MONTHLY PRICE
PASS	8 / 8	KRW 22K
Implemented and verified	Zero failures	Per workspace

01 / PROJECT OVERVIEW

Project overview

GearLedger lets a small creative workspace register equipment checkouts, work from an overdue-first return queue, and separate available, returned, and maintenance value without deploying a full asset-management suite.

02 / TARGET CUSTOMER

Target customer

Photo and video studios, music rooms, maker spaces, and small education labs with two to ten operators sharing valuable portable equipment.

03 / PROBLEM

Problem

When checkout details live in chat messages or memory, equipment becomes overdue, staff cannot tell what is available, and damaged items may be loaned again before inspection. The financial exposure stays invisible.

PRODUCT

Three focused jobs

04 / CORE FEATURES

Core features

The MVP implements one complete equipment return loop:

01	Validated equipment checkout Capture asset, category, borrower alias, value, due date, checkout status, and condition with strict limits.
02	Return urgency queue Order overdue, today, upcoming, maintenance, and returned records around the next operational action.
03	Value and condition control Total checked-out, maintenance, and returned asset value while keeping condition attached to every record.

05 / HOW TO USE

How to use

At handoff, register the asset with a non-identifying team or project alias and a return date. Process overdue and today items first. On return, set the condition to good, check, or repair; move unavailable equipment to maintenance before another checkout.

06 / SYSTEM ARCHITECTURE

System architecture

A static browser UI calls pure validation, urgency, and aggregation functions and persists state only in localStorage. Node tests exercise the same module. No account, backend, database, analytics, or external API is used.

Browser UI	Ledger module	Local storage	Node tests
HTML + CSS	Rules + totals	Device-only data	Pure functions

07 / TECHNOLOGY

Technology

Static HTML, responsive CSS, vanilla JavaScript, localStorage, bundled Noto Sans KR fonts, and the built-in Node.js test runner. Runtime dependencies: zero. Paid APIs: zero.

Captured from headless Chromium after a tripod checkout was validated and added.

08 / RUNNING SCREEN

The verified screen shows four checked-out assets, one overdue return, KRW 3.85M value out, and KRW 780K in maintenance after adding a KRW 320K tripod. Marking that tripod returned restored value out to KRW 3.53M and raised returned value to KRW 1.62M.

VERIFICATION

Evidence, not assumptions

09 / TEST RESULTS

Test results

Every required check was executed against the generated project.

Check	Result	Evidence
Dependency installation	PASS	npm install completed; zero runtime dependencies
Application start	PASS	README HTTP server returned status 200
Core workflow	PASS	Value out KRW 3.53M to 3.85M to 3.53M
Automated tests	8 / 8	Validation, urgency, totals, status, condition
README commands	PASS	Install, test, and start commands executed
Credential scan	PASS	No secret patterns and no .env file
PDF	PASS	Generated, parsed, rendered, visually inspected

10 / SECURITY AND LIMITATIONS

Security and limitations

GearLedger sends no data outside the browser and encourages team or project aliases rather than personal contact details. Text, value, date, status, and condition inputs are validated; displayed text is HTML-escaped. Device-local data is unencrypted and disappears when browser storage is cleared. The MVP has no barcode scanning, photo evidence, notifications, audit history, or multi-user access.

GO TO MARKET

Protect one asset before it goes missing

11 / PRICING AND REVENUE MODEL

Pricing and revenue model

Charge KRW 22,000 per workspace per month for local use and updates. Preventing one delayed production or identifying one maintenance issue early can exceed the annual subscription cost. The MVP itself has no hosting or paid-service dependency.

12 / FIRST CUSTOMER ACQUISITION

First customer acquisition

Recruit fifteen photo studios, rehearsal rooms, maker spaces, and school media labs through local operator communities. Load their ten most frequently shared items and run a two-week return-control trial. Convert by measuring overdue count, unavailable time, and value protected.

13 / IMPROVEMENT PLAN

Improvement plan

Priorities after observing real handoff routines:

- JSON backup and restore for device migration.
- Printable QR labels for faster local record lookup.
- Optional return reminders and condition-photo attachments.
- Encrypted sync, roles, and immutable audit history after team demand is proven.

PRODUCT BOUNDARY

GearLedger should remain an operational checkout tool. It should not store unnecessary personal data, claim to replace insurance or legal agreements, or expose a shared asset register without access control.

ASSESSMENT

Overall evaluation

14 / OVERALL SCORE

Overall score

Six dimensions are scored out of 10. A higher maintenance score means easier maintenance.

Dimension	Score	Rationale
Customer problem strength	8/10	Overdue or unavailable equipment directly disrupts paid work.
One-day completeness	9/10	The checkout, prioritize, return, and condition loop is tested.
Sales potential	8/10	Protected asset value gives a concrete subscription value case.
Maintenance difficulty	9/10	Zero runtime dependencies and no backend keep maintenance light.
Differentiation	8/10	Value exposure and condition status sharpen a generic checkout list.
Scalability	8/10	QR labels, photos, reminders, roles, and audit history can expand it.

OVERALL

8.3 / 10

Proceed to a fifteen-workspace
return-control trial.

Verdict: technically complete and visually polished for a business MVP. The next useful evidence is whether overdue count and unavailable equipment time fall during a real two-week trial.